

What are Real numbers?

Real numbers are the set numbers that do not include any imaginary value. It includes all the positive integers, negative integers, fractions, and decimal values. It is generally denoted by 'R'.

OR

All the negative and positive integers, decimal and fractional numbers without imaginary numbers are called real numbers.

Real numbers can be explained as the union of both rational and irrational numbers.

Examples

Rational numbers $\rightarrow \{5/3, 0.63, -6/5, 0.7116, \dots\}$

Irrational numbers $\rightarrow \{\sqrt{3}, \sqrt{5}, \sqrt{11}, \sqrt{21}, \pi(\text{Pi})\}$

Integers $\rightarrow \{-3, -2, -1, 0, 1, 2, 3, \dots\}$

Whole numbers $\rightarrow \{0, 1, 2, 3, 4, \dots\}$

Natural numbers $\rightarrow \{1, 2, 3, 4, \dots\}$

Types Of Numbers

There are different types of numbers categorized into sets by the real number system. The types are described below:

Rational numbers: Rational numbers are the numbers that can be expressed as the ratio of two integers. It includes all the integers and can be expressed in terms of fractions or decimals. It is denoted by 'Q'.

It can be expressed as $4/3, 0.83, -7/5, 0.711$

Irrational numbers: Irrational numbers are numbers that cannot be expressed in fractions or ratios of integers. It can be written in decimals and have endless non-repeating digits after the decimal point. It is denoted by 'P'.

It can be expressed as $\sqrt{5}, \sqrt{11}, \sqrt{21},$

Integers: Integers are the set of numbers including all the positive counting numbers, zero as well as all negative counting numbers which count from negative infinity to positive

infinity. The set doesn't include fractions and decimals. The set of integers is denoted by 'Z'.

The set of integers can be represented as $Z = \dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots$

Decimal numbers: Any numeral value that consists of a decimal point is a decimal number.

It can be expressed as 2.5, 0.567, 9.08 etc.

Whole numbers: Whole numbers are positive numbers including zero, which counts from 0 to infinity. Whole numbers do not include fractions or decimals. The set of whole numbers is represented by 'W'.

The set can be represented as $W = 0, 1, 2, 3, 4, 5, \dots$

Natural numbers: Natural numbers are the positive numbers that count from 1 to infinity. The set of natural numbers is represented by 'N'. It is the numbers we generally use for counting.

The set of natural numbers can be represented as $N = 1, 2, 3, 4, 5, 6, 7, \dots$

What are Properties of Real Numbers?

The Properties of Real Numbers are as follows:

- Closure Property
- Commutative Property
- Associative Property
- Distributive Property
- Identity Element Property
- Inverse Element Property

Closure Property

The closure Property of Real Numbers states that the addition and multiplication of any real number results in a real number. Let's consider the following examples for addition as well as multiplication.

Closure Property for Addition

$$2 + 5 = 7 \text{ and } \sqrt{2} + 5\sqrt{2} = 6\sqrt{2}$$

Where, 2 and 5 (both real numbers) added to get 7(real number) and $\sqrt{2}$ and $5\sqrt{2}$ (both real numbers) to get $6\sqrt{2}$ (real number).

Closure Property for Multiplication

$$6 \times 5 = 30 \text{ and } \sqrt{3} \times 2\sqrt{5} = 2\sqrt{(15)}$$

Where, 6 and 5 (both real numbers) multiply together to get 30(real number) and $\sqrt{3}$ & $2\sqrt{5}$ (both real numbers) are multiplied together to get $2\sqrt{(15)}$ (real number).

Commutative Property

This property states that the order of operation for addition and multiplication doesn't matter as it will give us the same result in any order we take the two values for which the operation is defined. In other words, if we have two real numbers a and b, then either we add them $a + b$ or $b + a$, the result remains the same. Similarly for multiplication, either ab or ba the result remains the same.

This property is only valid for addition and multiplication not for subtraction and division.

For addition $x + y = y + x$

Example: If we add 6 in 2 or add 2 in 6 results will be the same i.e.,

$$6 + 2 = 8 = 2 + 6$$

For multiplication $x \times y = y \times x$

Example: If we multiply both the real number (6 and 5) the results will be same i.e.,

$$6 \times 5 = 30 = 5 \times 6$$

Where, x and y are any two Real Numbers.

Associative Property

This property states that when three or more numbers are added or multiplied then the result is the same regardless of the grouping of the components.

The addition or multiplication in which order the operations are performed does not matter as long as the sequence of the numbers is not changed. This is defined as the associative property.

That is, rearranging the numbers in such a manner that will not change their value.

For addition $(x + y) + z = x + (y + z)$

Example: If we add 6 in 2 or add 2 in 6 results will be the same i.e.,

$$(4 + 5) + 6 = 15 = 4 + (5 + 6)$$

For multiplication $(x \times y) \times z = x \times (y \times z)$

Example: If we multiply both the real numbers (6 and 5) the results will be the same i.e.,

$$(2 \times 5) \times 6 = 60 = 2 \times (5 \times 6)$$

Where, x, y, and z are any three Real Numbers.

Distributive Property

This property helps us to simplify the multiplication of a number by a sum or difference. It distributes the expression as it simplifies the calculation.

- $x \times (y + z) = x \times y + x \times z$
- $x \times (y - z) = x \times y - x \times z$

Example: Simplify $4 \times (5 + 6)$

Solution:

$$4 \times (5 + 6) = 4 \times 5 + 4 \times 6$$

$$\Rightarrow 4 \times (5 + 6) = 20 + 24$$

$$\Rightarrow 4 \times (5 + 6) = 44$$

Identity Element Property

This is an element that leaves other elements unchanged when combined with them. The identity element for the addition operation is 0 and for multiplication is 1.

For addition, 0 is the identity element for the Real Numbers i.e.,

$$a + 0 = a = 0 + a$$

For multiplication, 1 is the identity element for the Real Numbers i.e.,

$$a \times 1 = a = 1 \times a$$

Note: 0 is the additive identity and 1 is the multiplicative identity.

Inverse Element Property

The inverse of an element is the number in the given set which when operated on the element gives the identity. In other words, in addition for any real number a , the inverse of a is such a number when added with results in identity (0) or for multiplication for any real number b , the inverse is such a real number when multiplied with b results in identity (1).

For addition: $a + (-a) = 0$, i.e., $-a$ and a are inverses of each other under Addition.

Example: Find the additive inverse of $1/5$.

Solution:

Let the inverse of $1/5$ be x , then using the property of inverse

$$1/5 + x = 0$$

$$x = -1/5$$

Similarly, 7 & -7 are the additive inverse of each other.

For Multiplication: $a \times 1/a = 1$ i.e., a and $1/a$ are inverse of each other under multiplication